higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

10 April 2026

This marking guideline consists of 6 pages.

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TVET Exam Papers NATED & NCV	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
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SECTION A

QUESTION 1: FLOW MEASUREMENT

1.1 1.1.1 Laminar flow: Smooth fluid flow that has a parabolic-shaped flow profile✓ with no mixing between the streamlines.✓

1.1.2 Turbulent flow: Fluid flow with a flattened parabola-shaped flow profile, the streamlines are not present,✓ and the fluid is freely intermixing.✓

1.1.3 Incompressible fluid: A fluid where there is very little volume change when subjected to a change in pressure.✓✓

1.1.4 Compressible fluid: A fluid where the volume and density change when subjected to a change in pressure.✓✓

(4 × 2) (8)

1.2 1.2.1 Drag-body flowmeter (1)

1.2.2 A = drag disc✓

B = strain gauge transducer✓ or force-measuring mechanism (2)

1.2.3 The meter consists of a blunt "paddle" (or "drag disk") inserted into the flow stream.✓ The force exerted on this disk by the moving fluid is sensed by a special transmitter mechanism,✓ which then outputs a signal corresponding to flow rate proportional to the square of fluid velocity.✓✓ (4)

1.3 1.3.1

$$Q = 3{,}33\ (L - 0{,}2\ H)H^{\frac{3}{2}}$$

$$Q = 3{,}33\ (3 - 0{,}2\ \times 4)4^{\frac{3}{2}}\checkmark$$

$$= 59{,}141\ m^3/hr\checkmark$$ (2)

1.3.2

$$Q = 3{,}37\ LH^{\frac{3}{2}}$$

$$Q = 3{,}37 \times 3 \times 4^{\frac{3}{2}}\checkmark$$

$$Q = 80{,}88\ m^3/hr\checkmark$$ (2)

1.3.3 Due to the contraction in the rectangular weir, the flow rate is less than that of the trapezoidal weir because there is less restriction to the flow. (1)

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QUESTION 2

2.1

- No energy is lost or gained in the venturi tube, or all energy is conserved.
- No mass is lost or gained in the venturi tube (all mass is conserved, or density remains constant).
- Fluid is incompressible.
- The venturi tube's centreline is level (no height changes to consider), or the height stays constant. (4)

2.2

Applying the last two assumptions to Bernoulli's equation, we see that the "elevation head" term drops out of both sides, since z, ρ, and g are equal at all points in the system: ✓

$$\frac{v_1^2\rho}{2} + P_1 = \frac{v_2^2\rho}{2} + P_2 \quad ✓$$

Now we will algebraically re-arrange this equation to show pressures at points 1 and 2 in terms of velocities at points 1 and 2:

$$\frac{v_2^2\rho}{2} - \frac{v_1^2\rho}{2} = P_1 - P_2 \quad ✓$$

Factoring $\frac{\rho}{2}$ out of the velocity head terms:

$$\frac{\rho}{2}(v_2^2 - v_1^2) = P_1 - P_2 \quad ✓$$

The Continuity equation shows us the relationship between velocities v_1 and v_2 and the areas at those points in the venturi tube, assuming constant density (ρ):

$$A_1 v_1 = A_2 v_2 \quad ✓$$

Specifically, we need to re-arrange this equation to define v_1 in terms of v_2 so we may substitute into Bernoulli's equation:

$$v_1 = \left(\frac{A_2}{A_1}\right) v_2 \quad ✓$$

Performing the algebraic substitution:

$$\frac{\rho}{2}(v_2^2 - \left[\left(\frac{A_2}{A_1}\right) v_2\right]^2) = P_1 - P_2 \quad ✓$$

Distributing the "square" power:

$$\frac{\rho}{2}(v_2^2 - \left(\frac{A_2}{A_1}\right)^2 v_2^2) = P_1 - P_2 \quad ✓$$

Factoring v_2^2 out of the outer parentheses set:

$$\frac{\rho v_2^2}{2}(1 - \left(\frac{A_2}{A_1}\right)^2) = P_1 - P_2 \quad ✓$$

Solving for v_2, step by step:

$$\frac{\rho v_2^2}{2} = \left(\frac{1}{1 - \left(\frac{A_2}{A_1}\right)^2}\right)(P_1 - P_2) \quad ✓$$

$$\rho v_2^2 = 2\left(\frac{1}{1 - \left(\frac{A_2}{A_1}\right)^2}\right)(P_1 - P_2)$$

$$v_2^2 = 2\left(\frac{1}{1 - \left(\frac{A_2}{A_1}\right)^2}\right)\left(\frac{P_1 - P_2}{\rho}\right) \quad ✓$$

$$v_2 = \sqrt{2}\frac{1}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}}\sqrt{\frac{P_1 - P_2}{\rho}} \quad ✓$$

(12)
[16]

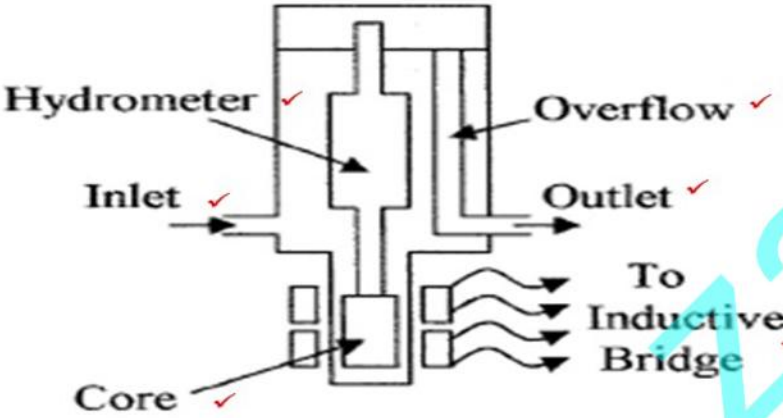
TOTAL SECTION A: 36

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SECTION B

QUESTION 3: MEASURING DENSITY, HUMIDITY, AND VISCOSITY

3.1 3.1.1



(6)

3.1.2
- A hydrometer with an iron core fitted to the lower side of the hydrometer is placed in a continuous flow of liquid.
- The liquid, with a specific density, moves the hydrometer to a certain level.
- The iron core will be positioned between the two coils, thus changing the inductance of the two coils accordingly.
- This change in inductance will be sent to an inductive bridge, which in turn will be sent to a recorder.
- The inductance will be directly proportional to the density of the liquid. (5)

3.2 3.2.1 1 = hair element;✓ 2 = arm linkage;✓ 3 = tension spring;✓ 4 = pointer;✓ and 5 = scale✓ (5)

3.2.2 Certain organic and synthetic fibres undergo changes in length✓ when exposed to moist atmosphere.✓ This change in length is mechanically amplified✓ and send to a read-out device.✓ (4)

3.2.3 It requires good circulation of the measured gas.✓ (1)

3.3 3.3.1 Viscosity is defined as the measure of a fluid's internal or intermolecular resistance to shear stress,✓✓ or simply the functional resistance which exists in flowing fluid. (2)

3.3.2 Stress is defined as force per unit area. (1)

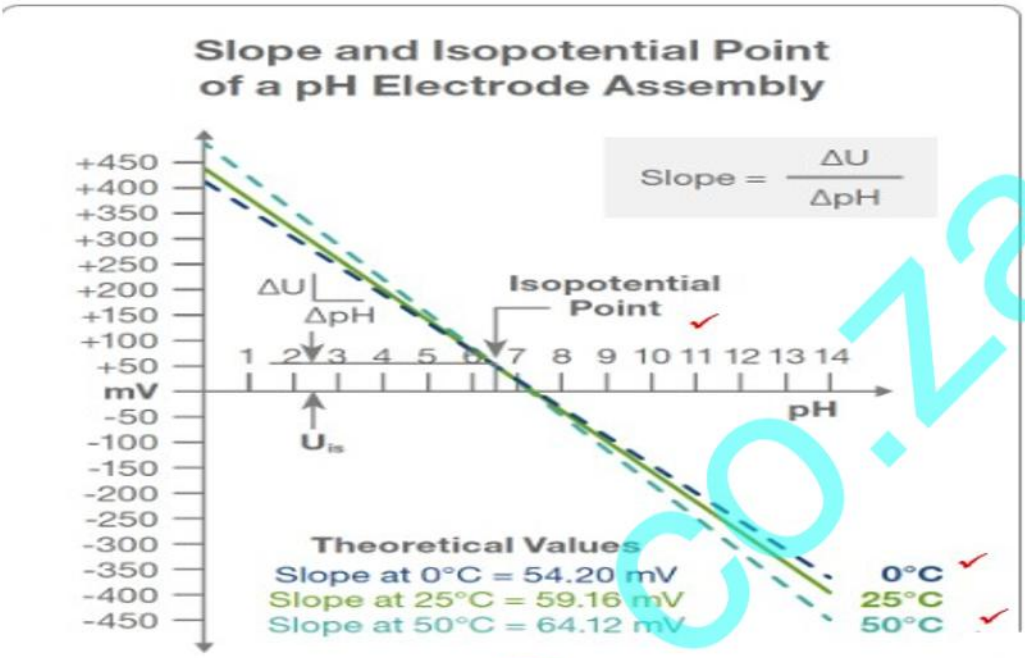
3.3.3 Stokes' law relates the viscosity of gases and liquids to the steady state velocity✓ of a known weight sphere falling through a fluid✓ of known density at a very low Reynolds number.✓ (3)

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QUESTION 4: MEASURING PH AND CONDUCTIVITY

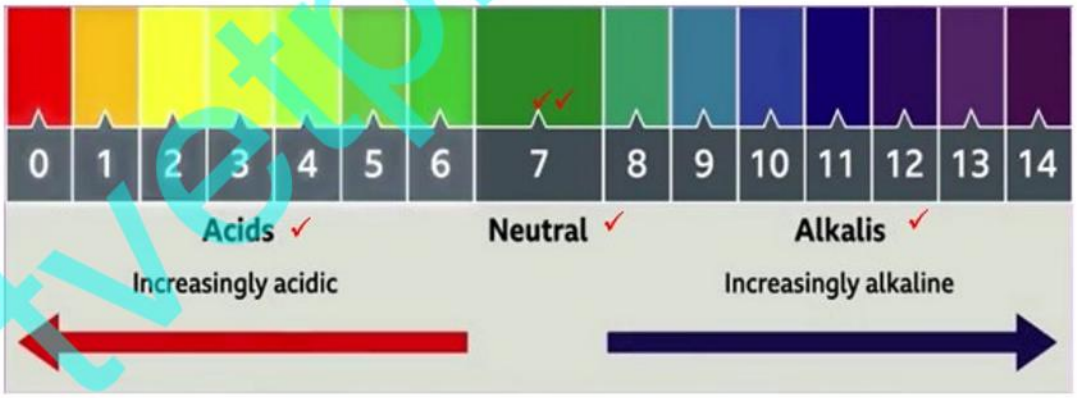
4.1 The isopotential point is the pH value at zero millivolt.✓✓ All temperature-dependent slope lines intersect the theoretical zero point (0 mV/pH 7).✓



(6)

4.2 The ideal housing must be corrosion-resistant,✓ and has strength against thermal✓ and mechanical shocks.✓ For measurement at high temperatures, it must be impermeable to water vapour.✓ (4)

4.3



(5)
[15]

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QUESTION 5: AUTOMATIC CONTROL

5.1 5.1.1 Proportional control (1)

5.1.2 Proportional control is defined as the action where the output of the controller is equal and opposite to the amount of deviation. (2)

5.1.3 The controlled variable oscillates around the set point✓ before it settles, leaving a permanent error called offset.✓ (2)

5.1.4 Offset is the difference between the set point and the controlled variable after the steady state condition is reached.✓✓ (2)

5.2 In an open-loop control, there is no feedback signal to the controller to reset the process.✓ Any deviation in the process must be corrected manually.✓ In a closed-loop control, there is a feedback mechanism which monitors the process variable.✓ Any deviation will be sensed and corrected automatically without human interaction.✓ (4)

5.3 5.3.1 A = step input
B = pulse input
C = ramp input
D = sinusoid input (4)

5.3.2 E = proportional✓ plus integral control✓
F = proportional✓ plus derivative control✓
C = proportional✓ plus integral✓ plus derivative control✓ (7)
[22]

TOTAL SECTION B: 64
GRAND TOTAL: 100

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